Exploring Bluez – 5.37

Table of Contents

Setup Details:

System configuration
Kernel : 3.18.0-9-generic with marvel patches
OS : Ubuntu 14.04.1 LTS
Linux Architecture : x86_64 GNU/Linux
RAM : 4GB

Installation:

1. In order to configure Bluez you need following software packages:
   1.1. Bluez (bluez-5.37.tar.xz)
   1.2. Udev library (udev-175.tar.gz)
   1.3. D-Bus library (dbus-1.8.6.tar.gz)
   1.4. GLib library (glib-2.38.2.tar.xz)
   1.5. readline (readline-6.3.tar.gz)
   1.6. libncurses (ncurses-5.9.tar.gz)
   1.7. libical (libical-1.0.1.tar.gz)
   1.8. libexpat (expat-2.0.1.tar.gz)
   1.9. libffi-dev (libffi-3.0.10.tar.gz)
   1.10. zlib library (zlib-1.2.8.tar.xz)

2. Downloading and installing all above packages:
   2.1. Download all above packages into /home/<user-name>/Bluez
   2.2. Now untar all the downloaded packages into /home/<user-name>/Bluez

   For Eg:

```
$mkdir /home/<user-name>/Bluez/
$ cd /home/<user-name>/Bluez/
$ tar -xvf /home/<user-name>/Bluez/bluez-5.37.tar.xz
$ tar -xvf /home/<user-name>/Bluez/glib-2.38.2.tar.xz
$ tar -xvf /home/<user-name>/Bluez/dbus-1.8.6.tar.gz
$ tar -xvf /home/<user-name>/Bluez/udev-175.tar.gz
$ tar -xvf /home/<user-name>/Bluez/zlib-1.2.8.tar.xz
$ tar -xvf /home/<user-name>/Bluez/libffi-3.0.10.tar.gz
$ tar -xvf /home/<user-name>/Bluez/libical-1.0.1.tar.gz
$ tar -xvf /home/<user-name>/Bluez/readline-6.3.tar.gz
$ tar -xvf /home/<user-name>/Bluez/expat-2.0.1.tar.gz
$ tar -xvf /home/<user-name>/Bluez/ncurses-5.9.tar.gz
```

   2.3. ------------------------Installation of Zlib----------------------------
   - cd /home/<user-name>/Bluez/zlib-1.2.8
   - ./configure --prefix=/usr/local/bluez/zlib-1.2.8
   - make
   - sudo make install

   2.4. -------------------------- Installation of Libffi---------------------------
   - cd /home/<user-name>/Bluez/libffi-3.0.10/
   - ./configure --prefix=/usr/local/bluez/libffi-3.0.10
   - make
   - sudo make install

   2.5. ---------------------- Installation of Expat -----------------------------
   - cd /home/<user-name>/Bluez/expat-2.0.1/
   - ./configure --prefix=/usr/local/bluez/expat-2.0.1/
   - make
   - sudo make install

   2.6. ----------------------Installation of Libical-----------------------------
   - sudo apt-get install cmake

- cd /home/<user-name>/Bluez/libical-1.0.1/
- cmake -DCMAKE_INSTALL_PREFIX=/usr/local/bluez/libical-1.0.1
- make
- sudo make install

2.7. -----------------Installation of Ncurses-----------------------------------

- cd /home/<user-name>/Bluez/ncurses-5.9
- ./configure --prefix=/usr/local/bluez/ncurses-5.9 --with-shared
- make
- sudo make install

2.8. -------------------Installing Readline ---------------------------------------

- cd /home/<user-name>/Bluez/readline-6.3/
- ./configure --prefix=/usr/local/bluez/readline-6.3 --with-curses
- make SHLIB_LIBS=/usr/local/bluez/ncurses-5.9/lib/libncurses.so
- sudo make install

2.9. ----------------- Installation of Glib --

- cd /home/<user-name>/Bluez/glib-2.38.2/
- ./configure --prefix=/usr/local/bluez/glib-2.38.2/ PKG_CONFIG_PATH=/usr/local/bluez/libffi-3.0.10/lib/pkgconfig:/usr/local/bluez/zlib-1.2.8/lib/pkgconfig
- make
- sudo make install

2.10. -------------------Installation of Dbus---

- cd /home/<user-name>/Bluez/dbus-1.8.6/
- ./configure --prefix=/usr/local/bluez/dbus-1.8.6/ PKG_CONFIG_PATH=/usr/local/bluez/glib-2.38.2/lib/pkgconfig CPPFLAGS=-I/usr/local/bluez/expat-2.0.1/include LDFLAGS=-L/usr/local/bluez/expat-2.0.1/lib
- make
- sudo make install

2.11. ------------------Installation of Udev---

- cd /home/<user-name>/Bluez/udev-175/
- ./configure --prefix=/usr/local/bluez/udev-175 PKG_CONFIG_PATH=/usr/local/bluez/glib-2.38.2/lib/pkgconfig --disable-gtk-doc-html --disable-logging --disable-gudev --disable-hwdb --disable-keymap --disable-mtd-probe --enable-rule-generator --disable-introspection
- make
- sudo make install

2.12. -------------------------Installation of BlueZ---------------------------------------

- $cd /home/<user-name>/Bluez/bluez-5.37/
- To enable BLE profile plugins and to build test scripts or tools which are under development, We need to configure with the following

options while installing Bluez

- $./configure --prefix=/usr/local/bluez/bluez-5.37 PKG_CONFIG_PATH=/usr/local/bluez/glib-2.38.2/lib/pkgconfig:/usr/local/bluez/dbus-1.8.6/lib/pkgconfig:/usr/local/bluez/udev-175/lib/pkgconfig:/usr/local/bluez/libical-1.0.1/lib/pkgconfig CPPFLAGS="-I/usr/local/bluez/readline-6.3/include -I/usr/local/bluez/udev-175/include" LDFLAGS=-L/usr/local/bluez/readline-6.3/lib --disable-systemd --enable-library --enable-experimental --enable-test --with-udevdir=/usr/local/bluez/udev-175

- (--enable-library install Bluetooth library
--enable-experimental enable experimental plugins (SAP,NFC, ...)
--enable-test enable test/example scripts)
- Build and install Bluez with make and make install commands.
- $make
- $sudo make install

---------------------- Congrats, You have installed Bluez now!!----------------------

2.13. If you get any installation errors, please follow below procedure

- $make distclean
- $./configure --prefix=/usr/local/bluez/bluez-5.37 PKG_CONFIG_PATH=/usr/local/bluez/glib-2.38.2/lib/pkgconfig:/usr/local/bluez/dbus-1.8.6/lib/pkgconfig:/usr/local/bluez/udev-175/lib/pkgconfig:/usr/local/bluez/libical-1.0.1/lib/pkgconfig CPPFLAGS="-I/usr/local/bluez/readline-6.3/include -I/usr/local/bluez/udev-175/include" LDFLAGS=-L/usr/local/bluez/readline-6.3/lib --disable-systemd --enable-library --enable-experimental --enable-test --with-udevdir=/usr/local/bluez/udev-175
- $make
- $sudo make install

Running Bluez

Before running installed Bluez, we need to check whether older version of Bluez is currently running or not. To check that use below command

$ ps -elf | grep "bluetoothd"

This older Bluez will start running automatically when system is up. To remove the conflict between the older Bluez and installed new Bluez, rename the older Bluez with some other name so that next time it will not run by default. To do that follow the below step:

$ sudo mv /usr/sbin/bluetoothd /usr/sbin/bluetoothd_4

Now to run our new Bluez, kill the older Bluez(/usr/sbin/bluetoothd) using using its pid. Find out the pid of older Bluez using below command

$ ps -elf | grep "bluetoothd"

Kill the bluetoothd using its pid. Use below command

$ sudo kill -9 <pid of the bluetoothd>

This is one time procedure need to be done when new Bluez is installed in customized path.

To run Installed Bluez follow the below steps:

1. Become a super user to run dbus and bluez
2. To run Bluez, first we need to run dbus
   check whether dbus-daemon is running or not by below command.
   # ps -elf | grep dbus-daemon
   if dbus-daemon is not running, then execute below command.
   #/usr/local/bluez/dbus-1.8.6/bin/dbus-daemon --system --nopidfile
3. Now check for the bluetoothd.
   check whether bluetoothd is running or not by below command.
   # ps -elf | grep bluetoothd
   if bluetoothd is not running, then execute below command.
   #/usr/local/bluez/bluez-5.37/libexec/bluetooth/bluetoothd -ndE --compat

   (-n, --nodetach Run with logging in foreground
   -d, --debug=DEBUG Specify debug options to enable
   -E, --experimental Enable experimental interfaces
   This -E will load the pluggings for BLE profile, NFC, etc.
)

To know more about the options use below command

/usr/local/bluez/bluez-5.37/libexec/bluetooth/bluetoothd --help

Profile Testing

1. Installation of Pulse Audio 6.0

1.1. Packages required

1.1.1. XML-Parser-2.41.tar.gz
1.1.2. intltool-0.50.2.tar.gz
1.1.3. libsndfile-1.0.26.tar.gz
1.1.4. alsa-lib-1.0.28.tar.bz2
1.1.5. json-c-0.11.tar.gz
1.1.6. sbc-1.3.tar.xz
1.1.7. libtool-2.4.3.tar.gz
1.1.8. pulseaudio-6.0.tar.gz

1.2. Download the above packages in the following path:-
/home/<UserName>/Downloads/pulseaudio

1.3. Untar the packages :-

1.3.1. tar -xvf XML-Parser-2.41.tar.gz
1.3.2. tar -xvf intltool-0.50.2.tar.gz
1.3.3. tar -xvf alsa-lib-1.0.28.tar.bz2
1.3.4. tar -xvf json-c-0.11.tar.gz
1.3.5. tar -xvf sbc-1.3.tar.xz
1.3.6. tar -xvf libsndfile-1.0.26.tar.gz
1.3.7. tar -xvf libtool-2.4.3.tar.gz
1.3.8. tar -xvf pulseaudio-6.0.tar.gz

1.4. Installation steps:-

1.4.1. ----------------------------XML-Parser----------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/XML-Parser-2.41
- perl Makefile.PL EXPATLIBPATH=/usr/local/bluez/expat-2.0.1/lib/ EXPATINCPATH=/usr/local/bluez/expat-2.0.1/include/
- make
- sudo make install

1.4.2. --------------------------intltool--------------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/intltool-0.50.2/
- ./configure
- make
- sudo make install

1.4.3. --------------------------libsndfile-------------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/libsndfile-1.0.26/
- ./configure --prefix=/usr/local/bluez/libsndfile-1.0.26 --disable-static
- make
- sudo make install
-

1.4.4. ---------------------------alsa-lib--------------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/alsa-lib-1.0.28/
- ./configure --prefix=/usr/local/bluez/alsa-lib-1.0.28

- make
- sudo make install
- sudo apt-get install doxygen
- make doc
- sudo install -v -d -m755 /usr/local/bluez/alsa-lib-1.0.28/share/doc/alsa-lib-1.0.28/html/search
- sudo install -v -m644 doc/doxygen/html/*.* /usr/local/bluez/alsa-lib-1.0.28/share/doc/alsa-lib-1.0.28/html
- sudo install -v -m644 doc/doxygen/html/search/* /usr/local/bluez/alsa-lib-1.0.28/share/doc/alsa-lib-1.0.28/html/search

1.4.5. -----------------------------json-c-----------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/json-c-0.11/
- ./configure --prefix=/usr/local/bluez/json-c-0.11 --disable-static
- make
- sudo make install

1.4.6. -----------------------------sbc--------------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/sbc-1.3/
- ./configure --prefix=/usr/local/bluez/sbc-1.3 --disable-static PKG_CONFIG_PATH=/usr/local/bluez/libsndfile-1.0.26/lib/pkgconfig
- make
- sudo make install

1.4.7. ------------------------------libtool----------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/libtool-2.4.3/
- ./configure --prefix=/usr/local/bluez/libtool-2.4.3
- make
- sudo make install

1.4.8. -----------------------------pulseaudio--------------------------------

- cd /home/<UserName>/Downloads/pulseaudio/pulseaudio-6.0
- ./configure --prefix=/usr/local/bluez/pulseaudio-6.0 --disable-bluez4 --enable-bluez5 --enable-alsa --enable-udev --without-caps PKG_CONFIG_PATH=/usr/local/bluez/json-c-0.11/lib/pkgconfig:/usr/local/bluez/libsndfile-1.0.26/lib/pkgconfig:/usr/local/bluez/alsa-lib-1.0.28/lib/pkgconfig:/usr/local/bluez/dbus-1.8.6/lib/pkgconfig:/usr/local/bluez/bluez-5.37/lib/pkgconfig:/usr/local/bluez/sbc-1.3/lib/pkgconfig:/usr/local/bluez/udev-175/lib/pkgconfig/ LDFLAGS=-L/usr/local/bluez/libtool-2.4.3/lib CPPFLAGS=-I/usr/local/bluez/libtool-2.4.3/include
- sudo apt-get install libbluetooth-dev
- make
- sudo make install

-------- Congrats you have installed pulseaudio-6.0 successfully ----------

Running Pulse Audio

Before running new pulseaudio, Please check whether pulseaudio is currently running or not using the following command.

$ps -elf | grep "pulseaudio"

If the executable is found then rename it with someother name using below command

$ mv /usr/sbin/pulseaudio /usr/sbin/pulseaudio-4.0

Then kill pulseaudio(i.e. /usr/sbin/pulseaudio) using kill command.

$ kill -9 <pid_of_pulseaudio>

This is one time procedure need to be done after pulseaudio is installed. Before running new pulseaudio, you have to run bluetoothd, else you will get error in pulseaudio saying " The name org.bluez was not provided by any .service files". Now Run pulseaudio in **Normal user** mode using below command.
$ /usr/local/bluez/pulseaudio-6.0/bin/pulseaudio -vvv

Note : Errros while running pulseaudio

Error	Solution
I: [pulseaudio] (alsa-lib)utils.c: could not open configuration file /usr/local/bluez/alsa-lib-1.0.28/share/alsa/ucm/HDA Intel/HDA Intel.con	sudo cp /usr/share/pulseaudio/alsa-mixer/profile-sets/extra-hdmi.conf /usr/local/bluez/pulseaudio-6.0/share/pulseaudio/alsa-mixer/profile-sets/

2. A2DP Profile Testing on Bluez-5.37

2.1. Run Bluez5.37

2.1.1. Need to be a **superuser** to run the daemon.

2.1.2. Check if installed dbus daemon is already running using below command.

- # ps -elf |grep dbus-daemon

(Note : In result, if you found '/usr/local/bluez/dbus-1.8.6/bin/dbus-daemon --system --nopidfile' then installed dbus is running. And if multiple are running kill them, make sure only one has to be running)

2.1.3. If it is not running, Run using below command

- # /usr/local/bluez/dbus-1.8.6/bin/dbus-daemon --system --nopidfile

2.1.4. Now run bluetoothd using below command
check whether bluetoothd is running or not by below command.
ps -elf | grep bluetoothd
if bluetoothd is not running, then execute below command.

- # /usr/local/bluez/bluez-5.37/libexec/bluetooth/bluetoothd -nd --compat

2.2. Run Pulseaudio

2.2.1. Run pulseaudio in normal user mode

2.2.2. Open a new terminal and run the command:
check whether pulseaudio is running or not by below command.
ps -elf | grep pulseaudio

if pulseaudio is not running, then execute below command.

- $ /usr/local/bluez/pulseaudio-6.0 /bin/pulseaudio –vvv

No	Error	Fix
1	D-Bus access denied	chang the permission from deny to allow in file /etc/dbus-1/system.d/bluetooth.conf in policy context. then cp /etc/dbus-1/system.d/bluetooth.conf /usr/local/bluez/dbus-1.8.6/etc/dbus-1/system.d/bluetooth.conf

Note : Once after copying the configure file, follow the below steps

1. To affect new modified configuration file, restart dbus
2. Kill current dbus and rerun it
3. Run bluetoothd and pulseaudio once again.

2.3. Run bluetoothctl

2.3.1. Run bluetoothctl in super user mode, use following command
/usr/local/bluez/bluez-5.37/bin/bluetoothctl

2.3.2. Currently we are running A2DP on Pulseaudio 6.0, this requires the source and sink to be paired before playing the audio using bluetoothctl .

2.4. Steps involved in pairing

2.4.1. power on
2.4.2. scan on
2.4.3. scan off after detecting the desired device (For Ex: Any phone that is acting as source in A2DP profile).
2.4.4. pair <BD-address>
2.4.5. connect <BD-address>

No	Error	Fix
1	Authentication failed while pairing using agent	changed the permission from deny to allow in the file /usr/local/bluez/dbus-1.8.6/etc/dbus-1/system.conf for the variable “send_type = method_call”.
2	If connect failed	kill the dbus-daemon, buetoothd,pulseaudio and bluetoothctl then run these all in same order

Note : Once after editing configure file, follow the below steps

1. To affect new modified configuration file, restart dbus
2. Kill current dbus and rerun it
3. Run bluetoothd and pulseaudio once again

2.5. Tried following things

a. Connected system to phone
b. Playing a song in phone as source audio is getting streamed to Linux machine.
c. Done the A2DP source and sink operations on two different system, by following below mentioned steps.

A2DP Source:

Aim : To Test A2DP Source role in Bluez-5.37 in different Bluez-5.37 machines/device

Steps invloved:

1. Ensure that both roles (A2DP source and A2DP Sink are supported) in Bluez-5.37

2. One can ensure it by running the command
   a) cd /home/<user_name>/bluez/bluez-5.37/tools
   b) sudo ./sdptool browse local

3. Now, one can see the list of services supported for Bluez-5.37
Scenario : If the user intended to get only A2DP Source role, execute the command
   a) cd /home/<user_name>/bluez/bluez-5.37/tools
   b) sudo **./sdptool del <Service RecHandle_for_A2DP_SNK>**
      Ex: ./sdptool del 0x1000f

4. After service deletion,
   4.1. Pre-requisites & Initial Connection Procedure

   a) Run bluetooth daemon and pulseaudio in background
   b) Run Bluetoothctl and give the command **scan on**
   c) wait until the BD_ADDRESS of the desired machine appears on the terminal
   d) Once the machine/device is detected, type command **scan off**
   e) Type commands **agent on** and **default agent**
   f) Type command **pair** <BD_ADDRESS>
   g) Type command **connect** <BD_ADDRESS>

   4.2 Test Procedure

   a) Play any audio from through any media players

   4.3 Expected Result & Observation

   a) Audio must be streamed to speakers/headphones on the established connection.
   b) One can check with the command **hcitool con**
   c) OR can check in hcidump using the command **hcidump -i <interface> -Xt**

A2DP Sink:

Aim : To Test A2DP Sink role in Bluez-5.37 in different Bluez-5.37 machines/device

Steps invloved:

1. Ensure that both roles (A2DP source and A2DP Sink are supported) in Bluez-5.37

2. One can ensure it by running the command
    a) cd /home/<user_name>/bluez/bluez-5.37/tools
    b) sudo ./sdptool browse local

3. Now, one can see the list of services supported for Bluez-5.37
Scenario : If the user intended to get only A2DP Sink role, execute the command
    a) cd /home/<user_name>/bluez/bluez-5.37/tools
    b) sudo **./sdptool del <Service RecHandle_for_A2DP_SOURCE>**
        Ex: ./sdptool del 0x1000e

4. After service deletion,
    4.1. Pre-requisites & Initial Connection Procedure

    a) Run bluetooth daemon and pulseaudio in background
    b) Run Bluetoothctl and give the command **discoverable on**
    c) Type commands **agent on** and **default agent**
    d) Type 'yes' for any authorization process for key or any services, when Bluetoothctl is expecting.

    4.2 Test Procedure

    a) Listen to audio from via speakers/headphones

    4.3 Expected Result & Observation

    a) Audio must be streamed to speakers/headphones on the established connection.
    b) One can check with the command **hcitool con**
    c) OR can check in hcidump using the command **hcidump -i <interface> -Xt**

3. HFP Profile testing on Bluez-5.37

3.1. Downloading ofono

3.1.1. https://www.kernel.org/pub/linux/network/ofono/
3.1.2. Download ofono-1.17.tar.xz version.
3.1.3. $ cd /home/<username>/Downloads
3.1.4. $ tar -xvf /home/<username>/ofono-1.17.tar.xz

3.2. Installing ofono

3.2.1. $ cd /home/<username>/ofono-1.17
3.2.2. $ ./configure --prefix=/usr/local/bluez/ofono-1.17 PKG_CONFIG_PATH=/usr/local/bluez/glib-2.38.2/lib/pkgconfig:/usr/local/bluez/dbus-1.8.6/lib/pkgconfig:/usr/local/bluez/udev-175/lib/pkgconfig
3.2.3. $ make
3.2.4. $ sudo make install

3.3. Working with ofono

3.3.1. Run Bluetoothd

3.3.1.1. Note : should be in super user

3.3.1.2. check whether dbus-daemon is running or not by below command.

ps -elf | grep dbus-daemon

if dbus-daemon is not running, then execute below command.

#/usr/local/bluez/dbus-1.8.6/bin/dbus-daemon --system --nopidfile

3.3.2. Run Bluetoothd

3.3.2.1. Note 1 : should be in super user

3.3.2.2. If Bluetoothd is not running, run it using below command

3.3.2.3. #/usr/local/bluez/bluez-5.37/libexec/bluetooth/bluetoothd -ndE

3.3.3. Run pulseaudio

3.3.3.1. Note 1 : Need to run in normal user mode

3.3.3.2. check whether pulseaudio is running or not by below command.

ps -elf | grep pulseaudio

if pulseaudio is not running, then execute below command.

$ /usr/local/bluez/pulseaudio-6.0/bin/pulseaudio -vvv

3.3.3.3. (Note : Before running pulseaudio, Make sure single dbus and bluetoothd is running)

3.3.4. Run ofono

3.3.4.1. Note 1 : should be in super user mode

3.3.4.2. check whether ofonod is running or not by below command.

ps -elf | grep ofonod

if ofonod is not running, then execute below command.

/usr/local/bluez/ofono-1.17/sbin/ofonod -nd

3.3.5. Run Bluetoothctl

3.3.5.1. Note 1 : Need to be in normal user mode

3.3.5.2. $ /usr/local/bluez/bluez-5.37/bin/bluetoothctl

3.3.5.3. [bluetooth]# power on

3.3.5.4. [bluetooth]# agent on

3.3.5.5. [bluetooth]# pair <bd-address>

3.3.5.6. [bluetooth]# connect <bd-address>

3.3.6. Run scripts to test HFP

3.3.6.1. Note : should be in super user mode

3.3.6.2. # export DBUS_SYSTEM_BUS_ADDRESS=unix:path=/usr/local/bluez/dbus-1.8.6/var/run/dbus/system_bus_socket

3.3.6.3. #cd /home/<username>/ofono-1.17/test

3.3.6.4. # ./enable-modem

3.3.6.5. # ./dial-number <write the number>

3.3.6.6. # ./answer-calls

3.3.7. To do HFP AG role, fallow the below steps :

HFP AG Role:
Aim : To Test HFP AG role in Bluez-5.37 and Ofono-1.17

i. Download & Install Phonesim (If you installed bluez from Makefile, then it is already installed)

Go to home folder::   cd /home/<user_name>
login as root user i.e. $ su
apt-get install git
git clone git://git.kernel.org/pub/scm/network/ofono/phonesim.git
apt-get install libqt4-dev
cd phonesim/
./bootstrap
./configure --prefix=/usr/local/bluez/phonesim
make
make install

ii. Pre-setup

$ sudo vim /usr/local/bluez/ofono-1.16/etc/ofono/phonesim.conf
In this file, Add these 4 lines

[phonesim]
Driver=phonesim
Address=127.0.0.1
Port=12345

iii. Run Ofonod (If you are executing ofonod by boot script, then there is no need of doing this step)

run ofonod in this way,

export OFONO_AT_DEBUG=1
/usr/local/bluez/ofono-1.17/sbin/ofonod -nd '*'
('*' for to see AT commands)

iv. Emulate pop-up window

The phonesim package has got default.xml file , we need to emulate this pop up window.

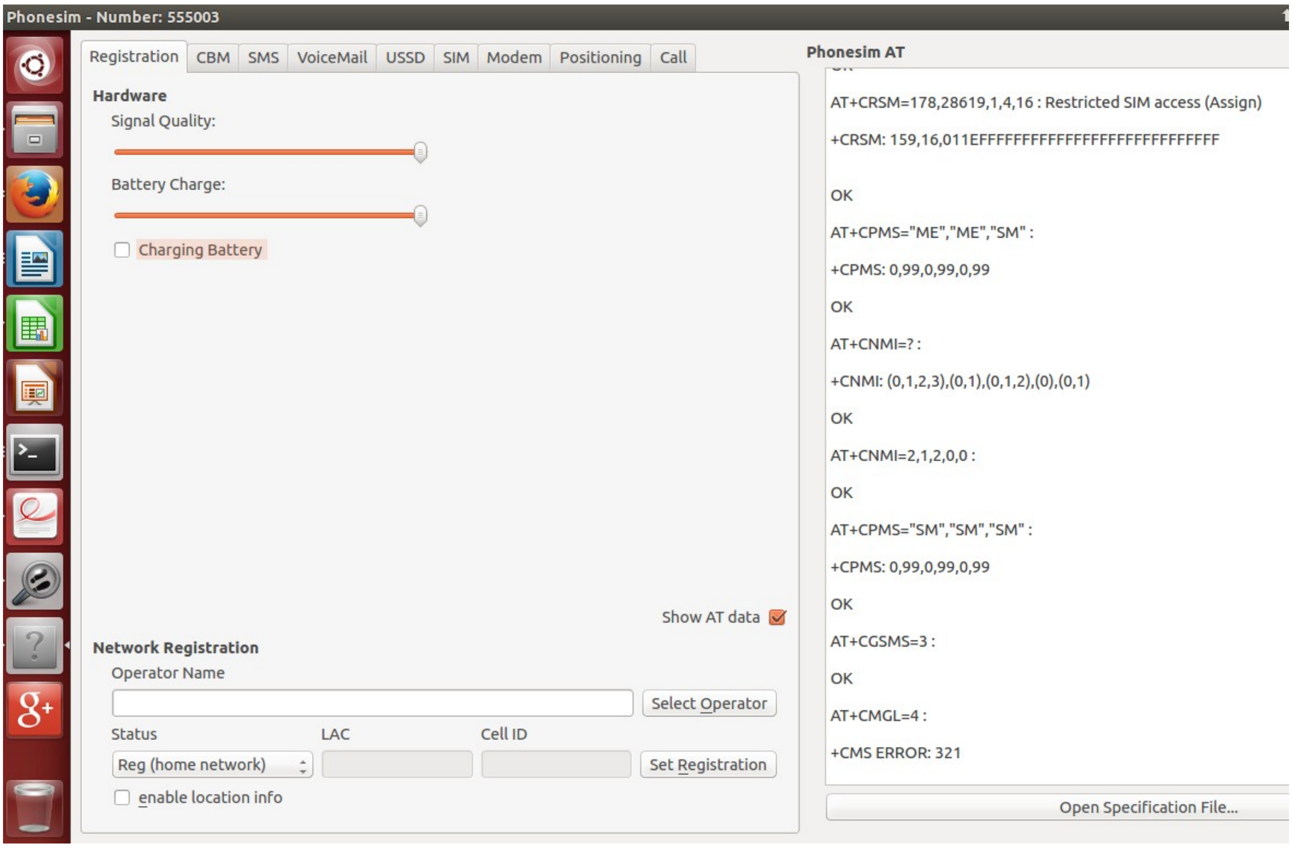
(In Bluez Installation Makefile, we already added phonesim installation steps. So if you installed ofono by Makefile then no need of doing step i, ii & iii)
cd ~/bluez/phonesim/src/
./phonesim -p 12345 -gui default.xml
(You will see nothing on the terminal. But, Don't worry!!)

v. Setproperty in Phonesim

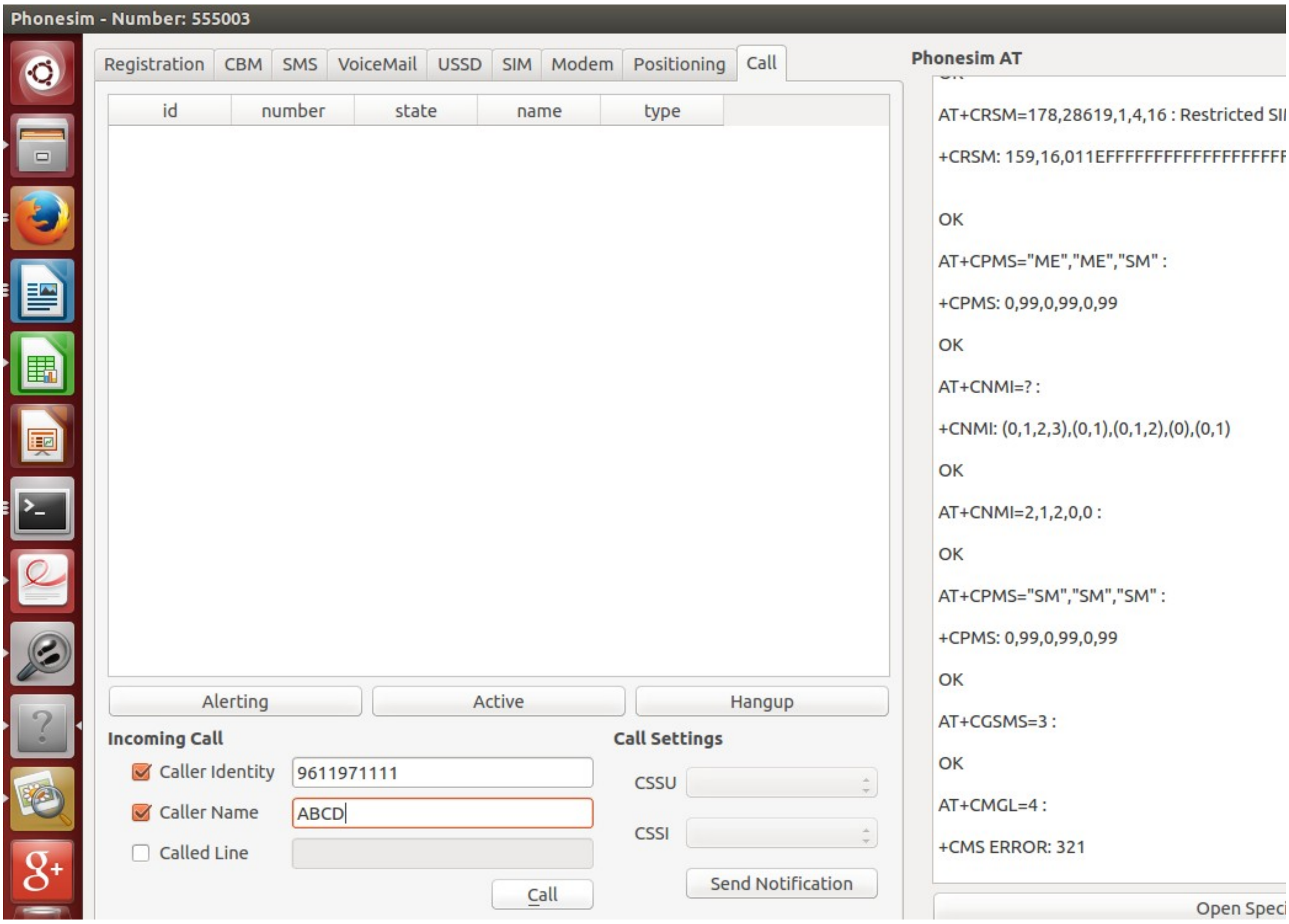
Go to customised dbus path:
i.e. # cd /usr/local/bluez/dbus-1.8.6/bin
then run,

./dbus-send --print-reply --system --dest=org.ofono /phonesim org.ofono.Modem.SetProperty string:"Powered" variant:boolean:"true"

Now, pop-up window is emulated



Now go to Call in the menu bar......



Type the number and name in the Caller identity and Caller name text box and then

Click on Call Button to simulate incoming call.

vi. Conclusion / Result
1. This enables to listen the incoming call that was simualted, into the headset I had connected via Bluetooth.
2. One can watch out ofonod logs for AT commands to Ringing, answering , dialling,etc

4. Message Access Profile Testing on Bluez-5.37

- Testing MAP in Bluez-5.37(system) with Android mobile(HTC)
- Mobile : Server which containing Messages
- Bluez-5.37(system) : client which access the messages

4.1. Make sure your bluetoothd is running

4.1.1. Become a super user
4.1.2. To run bluetoothd check whether dbus is already running using
4.1.3. #ps -elf | grep dbus-daemon
4.1.4. if the dbus is not running, run it using below command
4.1.5. #/usr/local/bluez/dbus-1.8.6/bin/dbus-daemon --system –nopidfile
4.1.6. check whether bluetoothd is running or not by below command.
ps -elf | grep bluetoothd
if bluetoothd is not running, then execute below command.
#/usr/local/bluez/bluez-5.37/libexec/bluetooth/bluetoothd -nd --compat

4.2. Make sure your Bluez system and mobile should be paired

4.2.1. Become a super user
4.2.2. Use bluetoothctl to pair with mobile
4.2.3. #/usr/local/bluez/bluez-5.37/bin/bluetoothctl
4.2.4. Follow below steps to get paired with your mobile in bluetoothctl
4.2.5. [bluetooth]# power on
4.2.6. [bluetooth]# agent on
4.2.7. [bluetooth]# default-agent
4.2.8. [bluetooth]# scan on
4.2.9. Once you find your device, do scan off
4.2.10. [bluetooth]# scan off
4.2.11. [bluetooth]# pair <BD_ADDRESS>
4.2.12. Once the pairing is successfull, then go for next step

4.3. Run obexd executable

4.3.1. Check were obexd is running or not using 'ps -elf | grep obexd'. If it is not there then execute below command in normal user.
4.3.2. $/usr/local/bluez/bluez-5.37/libexec/bluetooth/obexd -nd

4.4. Run scripts to test message access profile

4.4.1. To run script, you need to be normal user
4.4.2. Go to the directory of the test in Bluez
4.4.3. $cd /usr/local/bluez/bluez-5.37/lib/bluez/test
4.4.4. Run $./map-client --help to view the list of options to test MAP

4.5. Get the List of Messages from Mobile

4.5.1. Run following command inside test directory to view all messages
4.5.2. $./map-client -d <BD_ADDRESS> --lsmsg=LS_MSG
4.5.3. Allow MAP to access in DUT(Mobile) by pressing yes. On execution script will lists mobile messages in Bluez system.

4.6. List folders in current directory

4.6.1. Run following command to get list of folders in current directory
4.6.2. $ ./map-client -d <BD_ADDRESS> -l

4.7. Change the directory and list the folders within it
4.7.1. Run following command to view folders in changed directory
4.7.2. $./map-client -d <BD_ADDRESS> -c telecom/msg -l

4.8. List only Inbox messages.
4.8.1. Run below command to list only inbox messages.
4.8.2. $ ./map-client -d <BD_ADDRESS> -c telecom/msg/inbox --lsmsg=LS_MSG

4.9. You can do following operations
4.9.1. Get perticular message
4.9.2. Delete message
4.9.3. Mark Read/Unread status of message

Possible Errors while running the ./map-client script

dbus.exceptions.DbusException: org.bluez.obex.Error.Failed: Unable to find service record	There are two possibilities 1. Check Bluetooth is switched on in mobile 2. Mobile will not support MAP Note : Check the list of profiles mobile supports using sdptool application $sdptool browse <BD_ADDRESS>

5. Object Push Profile(OPP) Testing on Bluez-5.37:

- Testing OPP in Bluez-5.37(system) with Android mobile(HTC)
- Mobile : which recieves the files
- Bluez-5.37(system) : client which sends files

5.1. Make sure Bluetoothd is running (Refer sub 4.1 section)

5.2. Make sure your Bluez system and mobile should be paired (Refer sub 4.2 section)

5.3. Run obexd executable (Refer sub 4.3 section)

5.4. Run script to test object push profile
5.4.1. Go to the directory of the test in Bluez
5.4.2. $cd /home/<user_name>/bluez/bluez-5.37/test
5.4.3. Run $./opp-client --help to view the list of options to test OPP

5.5. Send file from Bluez system to Mobile
5.5.1. Create a file xxxx.txt in test folder with some content
5.5.2. Then run following command to send file
5.5.3. $./opp-client -d <BD_ADDRESS> -s xxxx.txt
5.5.4. Accept the file in mobile
5.5.5. Also we can send any .mp3 files from system to mobile

Note 1 : Before testing opp check whether mobile/tab supports it using sdptool.

Note 2 : We are running opp-client in source code test folder instead of /usr/local/bluez/bluez-5.37/lib/bluez/test/ because to copy file in this directory is not possible. It says Permission denied.

6. PBAP Testing on Bluez-5.37:

- Testing PBAP in Bluez-5.37(system) with Android mobile(HTC)
- Mobile : It acts as server which contains contacts
- Bluez-5.37(system) : client which recieves all the contacts

6.1. Make sure Bluetoothd is running (Refer sub 4.1 section)

6.2. Make sure your Bluez system and mobile should be paired (Refer sub 4.2 section)

6.3. Run obexd executable (Refer sub 4.3 section)

6.4. Run script to test Phone Book Access Profile

6.4.1. Go to the directory of the test in Bluez

6.4.2. $cd /usr/local/bluez/bluez-5.37/lib/bluez/test

6.4.3. Run $./pbap-client which help you to test PBAP

6.5. Get all the contacts from mobile to Bluez system

6.5.1. Run below command in test folder to get all contacts

6.5.2. $./pbap-client <BD_ADDRESS>

7. File Transfer Profile (FTP) Testing on Bluez-5.37:

- Testing FTP in Bluez-5.37(system) with Android mobile(Micromax Yureka)
- Mobile : Which recieves the files
- Bluez-5.37(system) : Which senda the files and removes them

7.1. Make sure Bluetoothd is running (Refer sub 4.1 section)

7.2. Make sure your Bluez system and mobile should be paired (Refer sub 4.2 section)

7.3. Run obexd executable (Refer sub 4.3 section)

7.4. Run script to test File Transfer Profile (FTP)

7.4.1. Go to the directory of the test in Bluez

7.4.2. $cd /home/<user_name>/bluez/bluez-5.37/test

7.4.3. Run $./ftp-client --help to view the list of options to test FTP

7.5. Transfer a file from Bluez system to mobile

7.5.1. Create a file xxxx.txt with some content in test directory

7.5.2. To send a file from Bluez system to Mobile, Use below command

7.5.3. $./ftp-client -d <BD_ADDRESS> -p xxxx.txt

7.6. List the Directories in current folder

7.6.1. To view the current directories in phone use below command

7.6.2. $./ftp-client -d <BD_ADDRESS> -l

7.7. Change the directory and push a file

7.7.1. Changing directory to PhoneMemory or Externel Memory and then push file

7.7.2. $./ftp-client -d <BD_ADDRESS> -c EXTERNAL_MEMORY -p xxxx.file

7.8. Change the directory and remove a file

7.8.1. To remove a file, Use below command

7.8.2. $./ftp-client -d <BD_ADDRESS> -c EXTERNAL_MEMORY -r xxxx.file

Note : Most of mobiles are not supporting this profile, We tested with Micromax Yureka.

8. HID Profile testing on Bluez-5.37

8.1. Pair the Bluez device with HID Device
8.2. Connect <HID_Device_BD_address>
8.3. Tested with Bluez-5.37 as HFP source and Speaker as HFP Sink.

Working of BLE Profiles

1. BLE Profile Configurations:

1.1. Run the bluetoothd with -E option
Ex. **bluetoothd -ndE**
check whether bluez is running or not by below command.
\# ps -elf | grep bluez
if 5 bluez processes is not running, then execute below command to run.

2. BLE profiles

2.1. Heart Rate Profile (HRP)
Bluez supports HRP collector role.

2.1.1. Test 1: We have tried Bluez as Collector role and CSR board as Heart Rate Sensor role. Heart Rate Sensor is sending the values and collector(Bluez) is able to receive and display the values. (CSR board is development kit in which HRP is implemented) We can also register HRP service using Bluez API's.

2.1.1.1. Run bluetooth daemon (path is /usr/local/bluez/bluez5.37/libexec/bluetooth then execute bluetoothd with option -ndE for LE profiles)

2.1.1.2. Run bluetoothctl application for pairing Collector and Sensor.

2.1.1.3. Run test scripts located in the path using --help option => /home/<user_name>/bluez-5.37/test (test folder from bluez source code)
Run ./test-heartrate –help
It will list out different usage options for the profile. Use the option according to usage.

2.1.2. Test 2: We have tried Bluez as Collector role and Polar Device as Heart Rate Sensor role. We are able to get the values from the polar device successfully.

2.1.2.1. Run bluetooth daemon (path is /usr/local/bluez/bluez-5.37/libexec/bluetooth then execute bluetoothd with option -ndE for LE profiles)

2.1.2.2. Run bluetoothctl application for pairing Collector and Sensor.

2.1.2.3. Run test scripts located in the path using --help option => /home/<user_name>/bluez-5.37/test (test folder from bluez source code)
It will list out different usage options for the profile. Use the option according to usage.

2.2. Cycling Speed and Cadence Profile(CSCP)
Bluez supports CSCP collector role.

We have tried Bluez as Collector role and CSR board as CSCP Sensor role. CSCP Sensor is sending the values and collector (Bluez) is able to receive and display the values. (CSR board is development kit in which CSCP is implemented to send some dummy values.)

2.2.1. Run bluetooth daemon (path is /usr/local/bluez/bluez-5.37/libexec/bluetooth then execute bluetoothd with option -ndE for LE profiles)

2.2.2. Run bluetoothctl application for pairing Collector and Sensor

2.2.3. Run test scripts located in the path using --help option => /home/<user_name>/bluez-5.37/test (test folder from bluez source code)
Run ./test-cyclingspeed –help
It will list out different usage options for the profile. Use the option according to usage.

2.3. Health Thermometer Profile (HTP)

Bluez supports CSCP collector role.
We have tried Bluez as Collector role and CSR board as HTP Sensor role. HTP Sensor is sending the values and collector (Bluez) is able to recieve and display the values. (CSR board is development kit in which HTP is imeplemented to send some dummy values.)

2.3.1. Run bluetooth daemon (path is /usr/local/bluez/bluez-5.37/libexec/bluetooth then execute bluetoothd with option -ndE for LE profiles)

2.3.2. Run bluetoothctl application for pairing Collector and Sensor.

2.3.3. Run test scripts located in the path using --help option => /home/<user_name>/bluez-5.37/test (test folder from bluez source code)
Run ./test-thermometer –help
It will list out different usage options for the profile. Use the option according to usage

2.4. Proximity Profile (PXP)

In PXP profile there are two roles.

- The Proximity Reporter - shall be a GATT server.
- The Proximity Monitor - shall be a GATT client.

As per the bluez documentation, it supports both the above mentioned roles. But in bluez 5.37, only the proximity monitor is working fine. There are some issues in bluez-5.37, because of which gatt-server is failing to start. So we tried with lower version 5.28, where the gatt server(Proximity Reporter) is working fine and also able to discover services.
Error :
bluetoothd[5879]: Error adding Link Loss service
bluetoothd[5879]: Not enough free handles to register service
bluetoothd[5879]: Not enough free handles to register service

2.4.1. Bluez-5.37 is working as a Proximity Monitor

We tested Proximity Profile (PXP) with Bluez-5.37 as proximity monitor role and TrackeR Device as Proximity Reporter role. Bluez is successfully monitoring the services like Link Loss Alert Level and Immediate Alert Level.

2.4.1.1. Run bluetooth daemon (path is /usr/local/bluez/bluez-5.37/libexec/bluetooth then execute bluetoothd with option -ndE for LE profiles)

2.4.1.2. Run bluetoothctl application for pairing Collector and Sensor.

2.4.1.3. Run test scripts located in the path using --help option => /home/<user_name>/bluez-5.37/test (test folder from bluez source code)
Run ./test-proximity –help

It will list out different usage options for the profile. Use the option according to usage.

Summary:

Profiles	Device Model (Tested With)	Tools Version (Dependency)	Comments
A2DP	HTC Desire 620G	Pulseaudio 6	Working: Audio streaming phone as source and system as sink
HFP	HTC Desire 620G	Pulseaudio 6 Ofono1.6	Working: Dialing a number Answering a call Rejecting a call
OPP	HTC Desire 620G	Obexd	Working: Sending .txt and MP3 files
PBAP	HTC Desire 620G	Obexd	Working: Accessing all the contacts from phone to system
FTP	Micromax Yureka	obexd	Tested and Working Sending files from Bluez Removing files in Mobile

Q & A :

1. Sometime during pairing/connection Bluez throw error:
   Need to register agent before pairing. Follow below Steps:
   [bluetooth]# agent on
   [bluetooth]# default-agent
   after follow the same pairing procedure.

2. Sometimes Inquiry does not happen and no response is seen.
   The process of scan should happen below
   [bluetooth]# scan on
   [bluetooth]# scan off
   If we try to execute “scan on” or “scan off” multiple time without above sequence it will give up error as “Failed to stop discovery: org.bluez.Error.Failed”.

3. Sometimes device shows connected bluez as well as phone but actually it not listed in the connected devices.
   Required more info, like Bluez support list of devices “[bluetooth]# devices” and list of paired devices “[bluetooth]# paired-devices” commands. Which list accordingly.

4. ./enable-modem throughs dbus error : modem_enable_error.txt (logs attached)
   Please check the following step before executing commands from other folders
   # export DBUS_SYSTEM_BUS_ADDRESS=unix:path=/usr/local/bluez/dbus-1.8.6/var/run/dbus/system_bus_socket (need to be executed in root user)
   Because by default system dbus path will be consider, hence we need to export the bluez dbus path while executing from other folder

5. Cannot make bluez as OPP Server: As such there are no separate steps mentioned for making Bluez as OPP server.
   Currently we don't have any OPP server script file, We are using Smartphone as server to test.
   And we are working on server role in perl.

6. Cannot LE advertise (even tried with shell scripts): using hcitool commands
   Please execute below hci commands.
   We need to set advertisement data, advertisement interval after we need to advertise.
   Details regarding ocf/ogf can be found in bluetooth sig file.
   # stop adverising
   hcitool -i hci0 cmd 0x08 0x000a 0x00

   # step up advertising interval
   hcitool -i hci0 cmd 0x08 0x0006 A0 00 A0 00 00 00 00 00 00 00 00 00 00

07 00

Step up scan responser data
hcitool -i hci0 cmd 0x08 0x0008 0A 02 01 00 03 03 0D 18 02 0A 0A 00 00 00 00 00 00 00 00 00 00 00

#step up advertisement data
hcitool -i hci0 cmd 0x08 0x0009 0B 0A 09 53 65 65 74 68 61 72 61 6d | 0F 0E 09 47 45 53 2d 48 65 61 72 74 52 61 74 65

start adverising
hcitool -i hci0 cmd 0x08 0x000a 0x01

7. Bluez crash was observed for OPP + HID + OPP scenario :opp+hid+opp.txt (logs attached)
   We need more information.
   Usually Bluez try to auto connect HID Ble devices. In that scenario if the pairing info is removed, master will try to connect continuously with slave device. Hence We need to remove the previously paired info from bluez and need to connect/pair freshly.